

1 Tests based on General Score Function

Consider the situation where $X_1, \dots, X_{n_1} \sim F(x - \mu_1)$ and $Y_1, \dots, Y_{n_2} \sim F(y - \mu_2)$ with $\mu_2 - \mu_1 = \Delta$. A test needs to be developed for the hypotheses $H_0 : \Delta = 0$ versus $H_A : \Delta > 0$. Under H_0 , the X s and Y s are identically distributed.

General scores are discussed in Section 2.5 of Hettmansperger and McKean (2011). A set of rank-based scores is generated by a function $\phi(u)$ defined on the interval $(0, 1)$. We assume that $\phi(u)$ is square-integrable and, without loss of generality, standardize as $\int_0^1 \phi(u) du = 0$ and $\int_0^1 \phi^2(u) du = 1$. The generated scores are then $a_\phi(i) = \phi[i/(n+1)]$, where $n = n_1 + n_2$. The Wilcoxon score function is given by $\phi_W(u) = \sqrt{12}[u - (1/2)]$.

For general scores, we define the process

$$S_\phi(\Delta) = \frac{1}{n} \sum_{j=1}^{n_2} a_\phi[R(Y_j - \Delta)]$$

where $R(Y_j - \Delta)$ denotes the rank of $Y_j - \Delta$ among $X_1, \dots, X_{n_1}$ and $Y_1 - \Delta, \dots, Y_{n_2} - \Delta$. A test statistic for the hypotheses $H_0 : \Delta = 0$ versus $H_A : \Delta > 0$ is $S_\phi(0)$.

It then follows that S_ϕ is distribution-free. Although exact null distributions of S_ϕ can be numerically generated, usually the null asymptotic distribution is used. The mean and variance of $S_\phi(0)$ under the null hypothesis are $\bar{a} = \frac{1}{n} \sum_{i=1}^n a_\phi(i)$ and $n_1 n_2 / n(n-1) \sum_{i=1}^n (a_\phi(i) - \bar{a})^2$.

2 Scale Problem

Besides differences in location, we are often interested in the difference between scales for populations. Let $X_1, \dots, X_{n_1}$ be a random sample with the common pdf $f[(x - \theta_1)/\sigma_1]$ and $Y_1, \dots, Y_{n_2}$ be a random sample with the common pdf $f[(y - \theta_2)/\sigma_2]$, where $f(x)$ is a pdf and $\sigma_1, \sigma_2 > 0$. In this section our hypotheses of interest are $H_0 : \eta = 1$ versus $H_A : \eta \neq 1$ where $\eta = \sigma_2/\sigma_1$. Besides discussing rank-based tests for these hypotheses, we also consider the associated estimation of η , along with a confidence interval for η . So here the location parameters θ_1 and θ_2 are nuisance parameters.

As discussed in Section 2.10 of Hettmansperger and McKean (2011), there are asymptotically distribution-free rank-based procedures for this problem. We discuss the Fligner–Killeen procedure based on folded, aligned samples. The aligned samples are transformed data defined by

$$X_i^* = X_i - \text{med}X, \quad i = 1, \dots, n_1$$

$$Y_j^* = Y_j - \text{med}Y, \quad j = 1, \dots, n_2.$$

A test statistic is $S = \sum_{j=1}^{n_2} R[|Y_j^*|]$ where the ranks are over the combined folded, aligned samples. Thus we can carry out a Wilcoxon rank sum test over the transformed data.

A more common test is discussed in Fligner and Killeen (1976) [FK] is obtained by using the normal score function.

$$\phi_{FK} = \left(\Phi^{-1} \left(\frac{u+1}{2} \right) \right)^2$$

where Φ is the standard normal cdf. This leads to the test statistic

$$S_{FK} = \sum_{j=1}^{n_2} \left(\Phi^{-1} \left(\frac{R|Y_j^*|}{2(n+1)} + \frac{1}{2} \right) \right)^2$$

Example (Effect of Ozone on Weight of Rats). Draper and Smith (1966) present an experiment on the effect of ozone on the weight gain of rats. Two groups of rats were selected. The control group ($n_1 = 21$) lived in an ozone-free environment for 7 days, while the experimental group ($n_2 = 22$) lived in an ozone environment for 7 days. At the end of the 7 days, the gain in weight of each rat was taken to be the response. For this example, the following code segment computes the Fligner–Killeen procedure for a two-sided alternative. The data are in the dataset `sievers`. We first split the groups into two vectors `x` and `y` as input to the function `fk.test`.

```
> data(sievers)
> x <- with(sievers,weight.gain[group=='Control'])
> y <- with(sievers,weight.gain[group=='Ozone'])
> fk.test(x,y)
statistic = 2.095976 , p-value = 0.03608434
95 percent confidence interval:
1.002458 5.636436
Estimate: 2.377034
```